



VIT[®]

Vellore Institute of Technology

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Fall Semester 2025-26

Lab Assignment – 4

Slot: L13+L14

Class: VL2025260105679

Branch: B.tech CSBS

Course code & title: CBS3005

Cloud, Microservices and Applications

Faculty name: Nithya K

DA by:

Kartikey Gupta

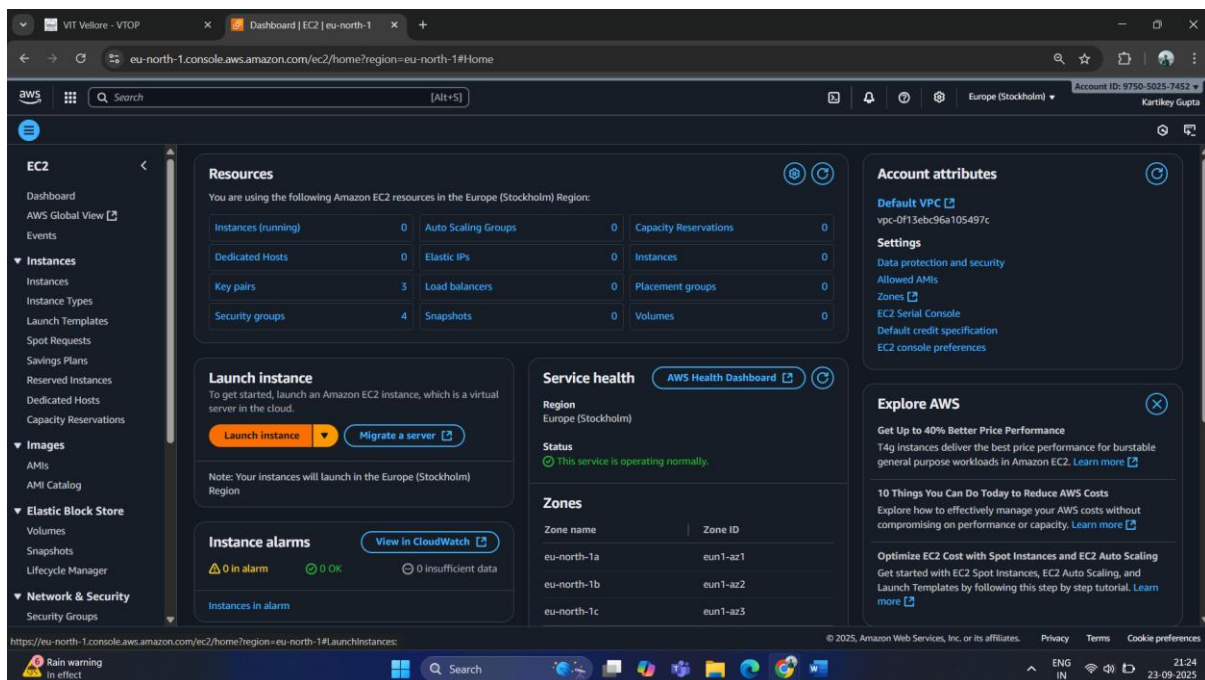
22BBS0105

1) A company wants to deploy a secure, scalable, and highly available web application on AWS for global users. Perform the following tasks in AWS and submit screenshots of each step as evidence:

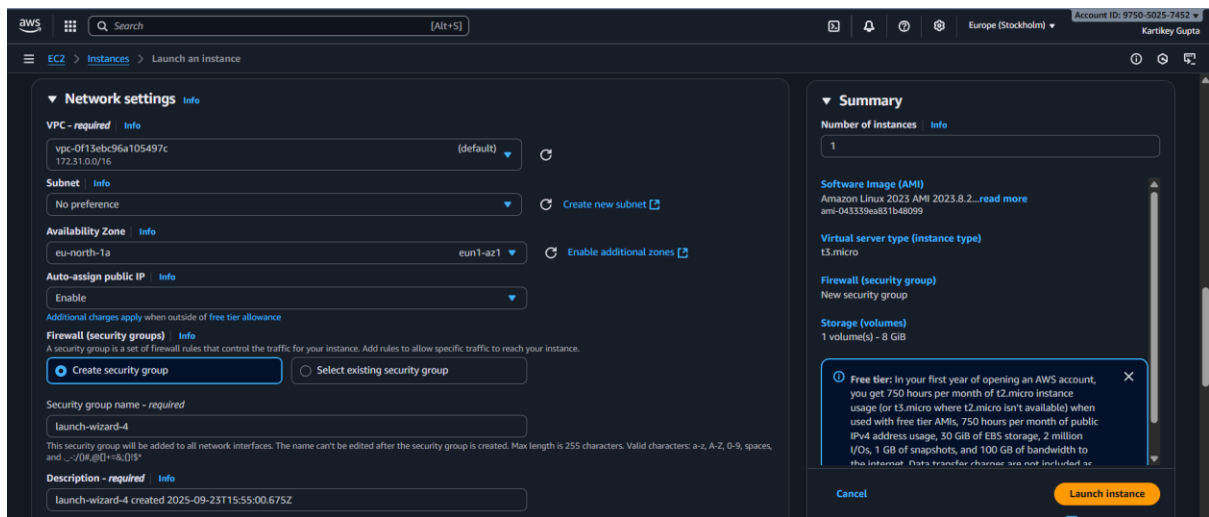
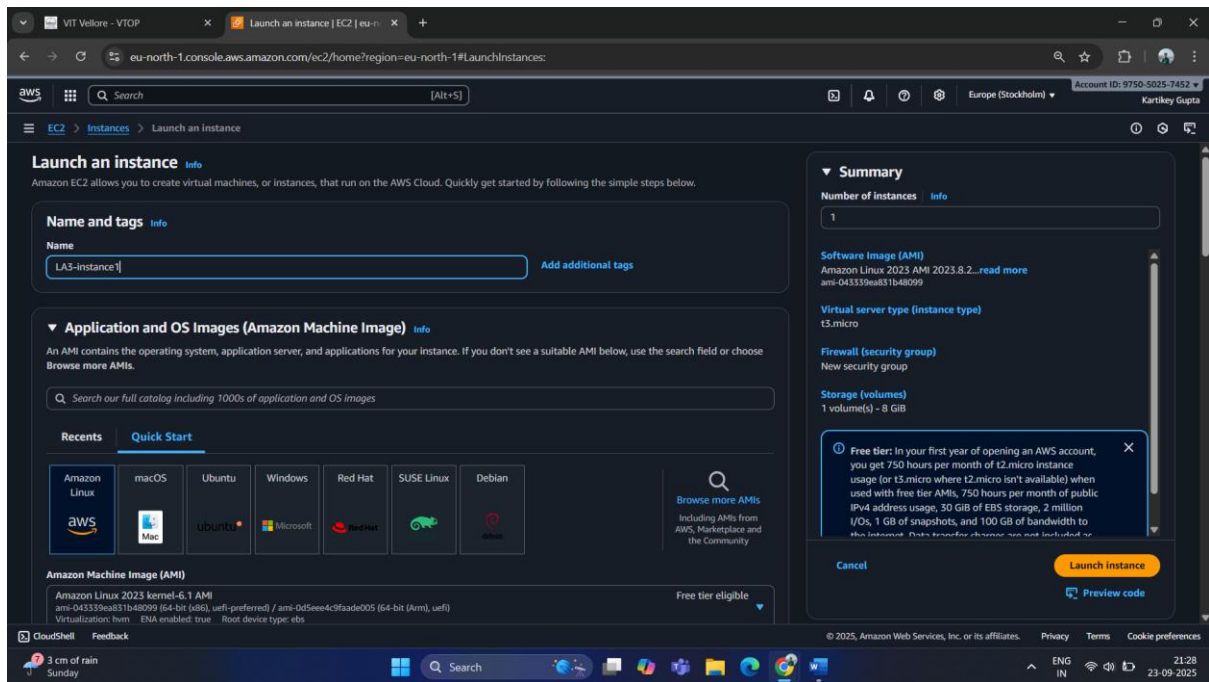
- (i) Launch multiple EC2 instances (web servers) and configure them in different Availability Zones.
- (ii) Create an Application Load Balancer (ALB) to distribute traffic across these instances.
- (iii) Configure health checks so that faulty instances are automatically removed from load balancing.
- (iv) Enable Auto Scaling to add/remove instances based on traffic demand.
- (v) Configure path-based routing: /auth requests go to the authentication service, /order requests go to the order processing service. Integrate the Load Balancer with Route 53 so that global users are routed to the nearest AWS region.

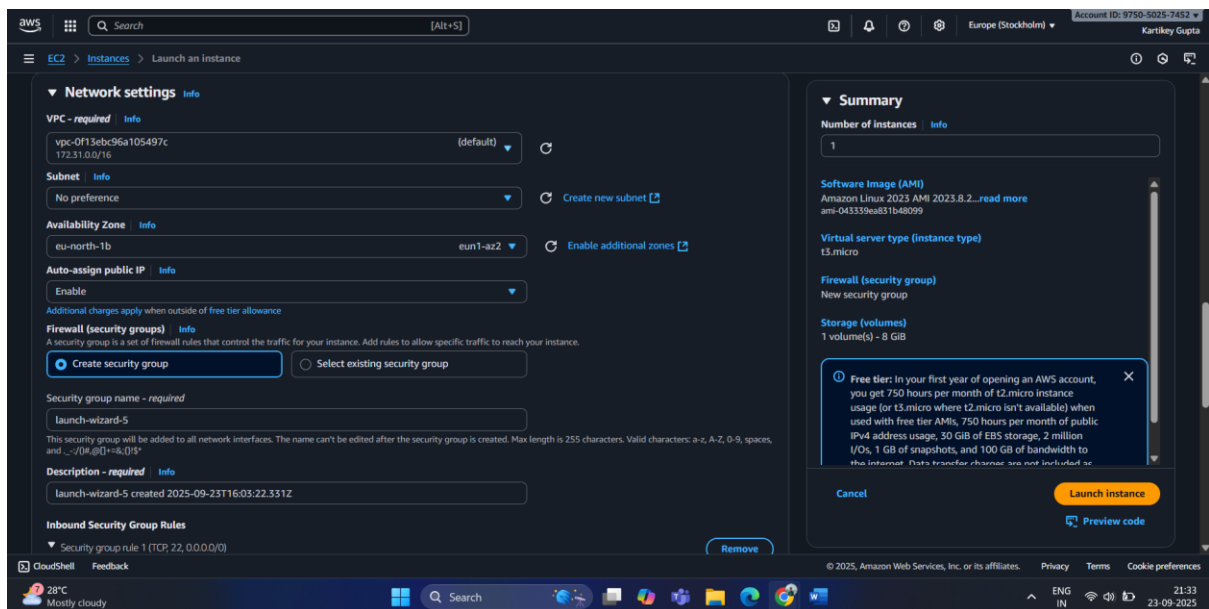
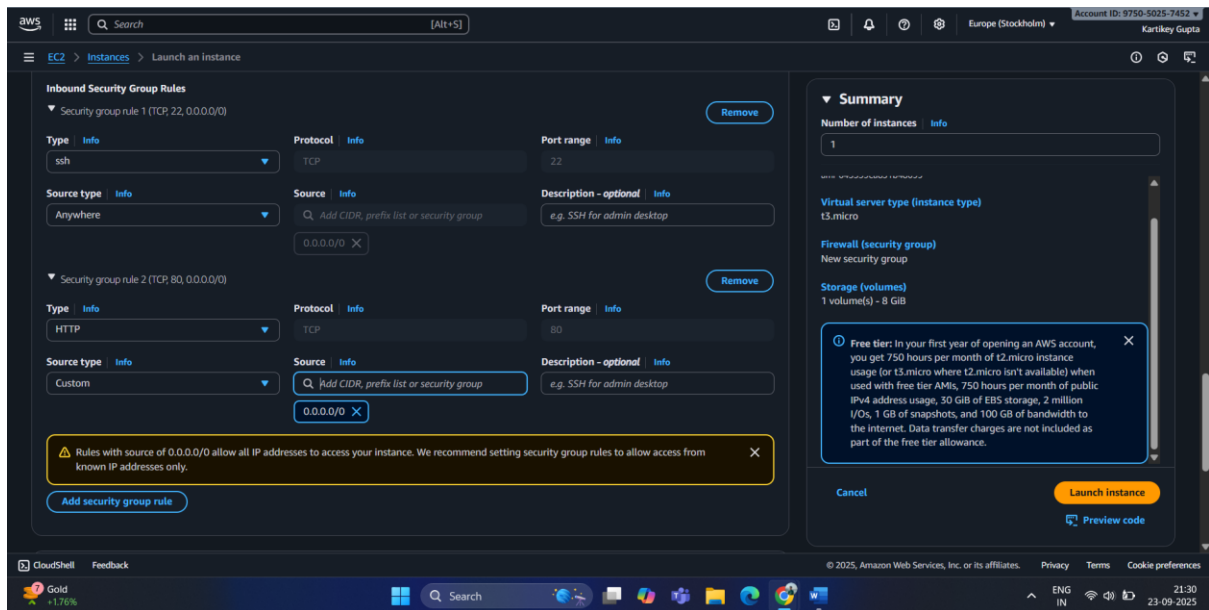
(i) Launch Multiple EC2 Instances in Different Availability Zones

1. Log in to AWS Management Console → go to **EC2**.

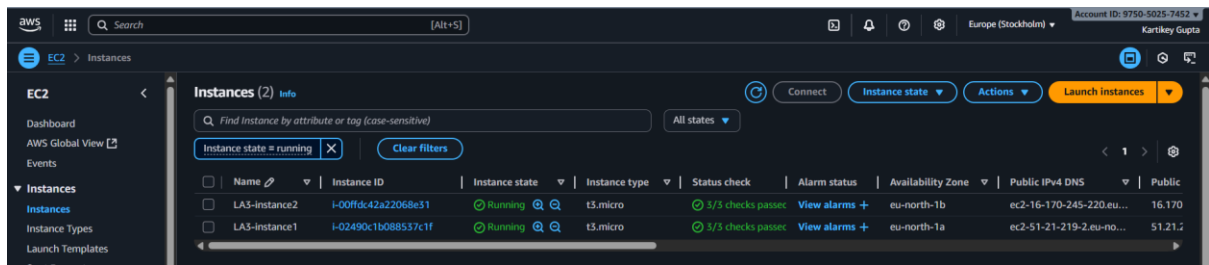


2. Click **Launch Instance** → choose Amazon Linux 2 / Ubuntu AMI.





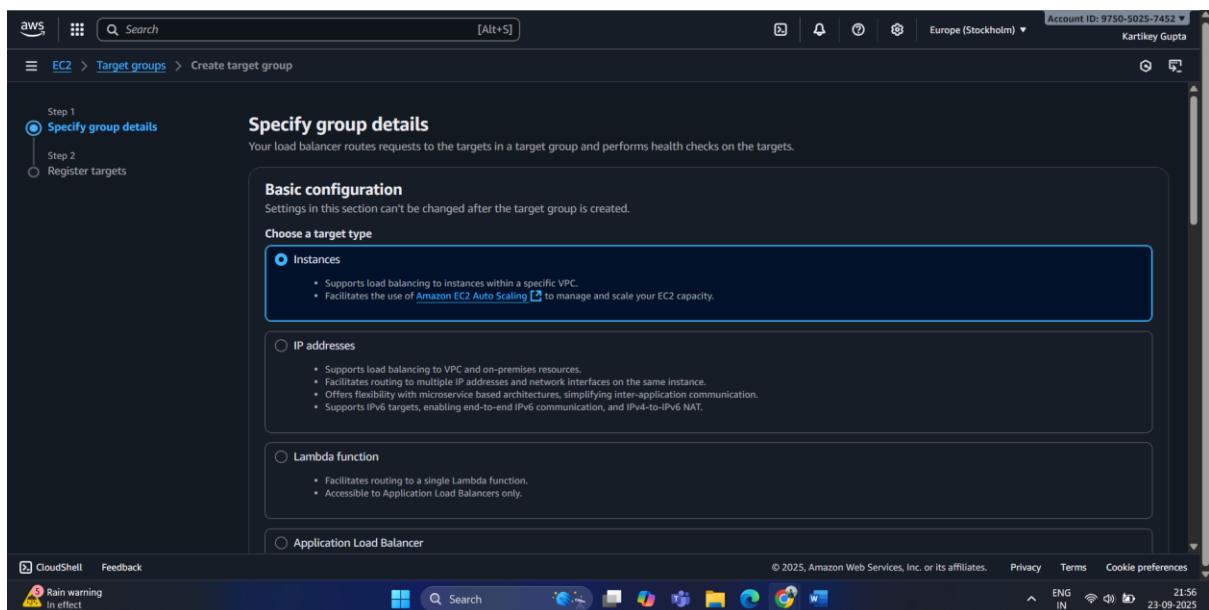
3. Select an **instance type** (e.g., t2.micro for free tier).
4. In **Network settings**:
 - Choose a **VPC** (default or custom).
 - Place **Instance 1** in Availability Zone A (e.g., ap-south-1a).
 - Place **Instance 2** in Availability Zone B (e.g., ap-south-1b).
5. Configure **Security Group**:
 - Allow HTTP (port 80) and SSH (port 22).
6. Launch instances and note **public IPs**.



(ii) Create Application Load Balancer (ALB)

Step 1: Create Target Group(s) first

1. Go to **EC2** → **Target Groups** → **Create target group**.
2. Configure:
 - **Target type:** Instances
 - **Name:** e.g., MyApp-TG
 - **Protocol:** HTTP
 - **Port:** 80
 - **VPC:** Select the same VPC where your EC2 instances are launched
 - **Health check:** HTTP, path /



Target group name
MyApp-TG
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol
Protocol for load balancer-to-target communication. Can't be modified after creation.
HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.
☒ IPv4
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
☐ IPv6
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
vpc-0f13ebc96a105497c (default) [Create VPC](#)

Protocol version
☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/
Up to 1024 characters allowed.

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with commas)
Include as pending below
2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2) [Remove all pending](#)

Filter targets ☐ Show only pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID
i-00ffdc42a22068e31	LA3-instance2	80	Running	launch-wizard-5	eu-north-1b	172.31.32.212	subnet-07a84a48af02e
i-02490c1b088537c1f	LA3-instance1	80	Running	launch-wizard-4	eu-north-1a	172.31.23.94	subnet-0f44bdb97830f

2 pending

[Cancel](#) [Previous](#) [Create target group](#)

3. Click **Next** and **register your EC2 instances** in this Target Group.

4. Complete Target Group creation.

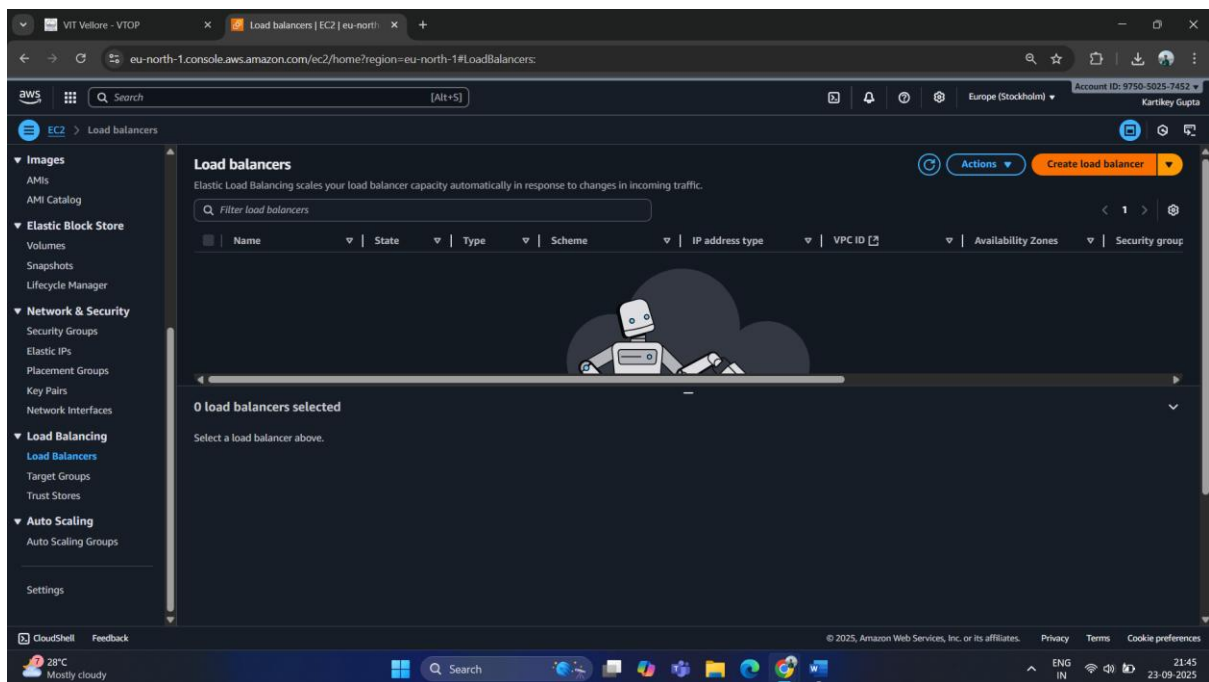
5. Install servers

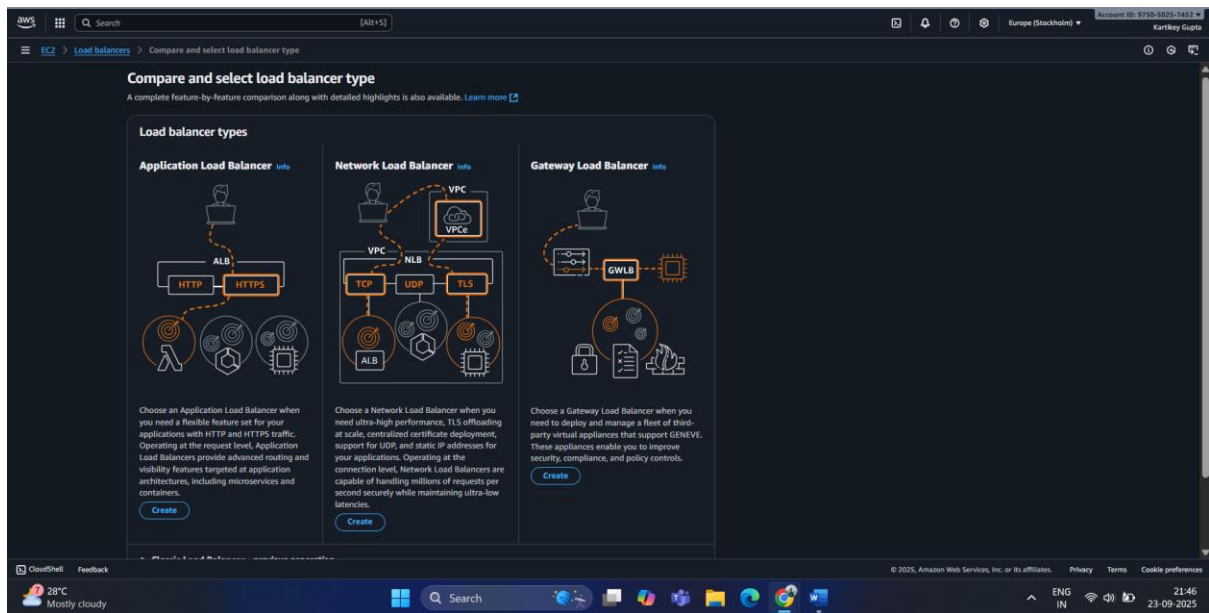
- sudo yum update -y
- sudo yum install -y httpd
- sudo systemctl start httpd

- `sudo systemctl enable httpd`
- `echo "Hello from $(hostname -f)" > /var/www/html/index.html`

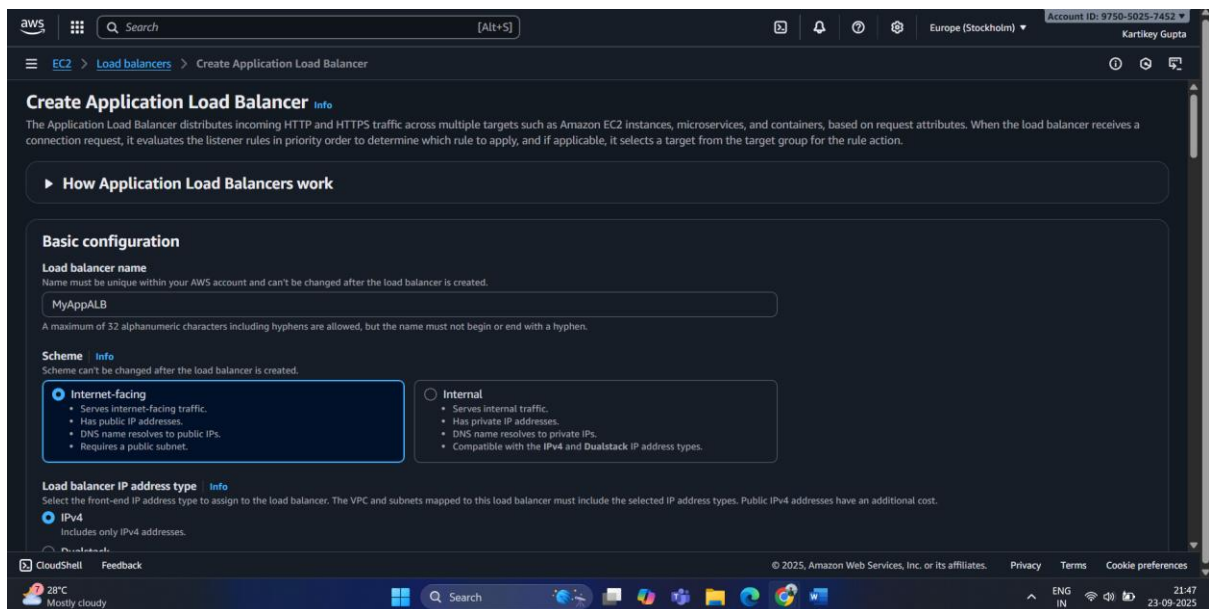
Step 2: Create ALB

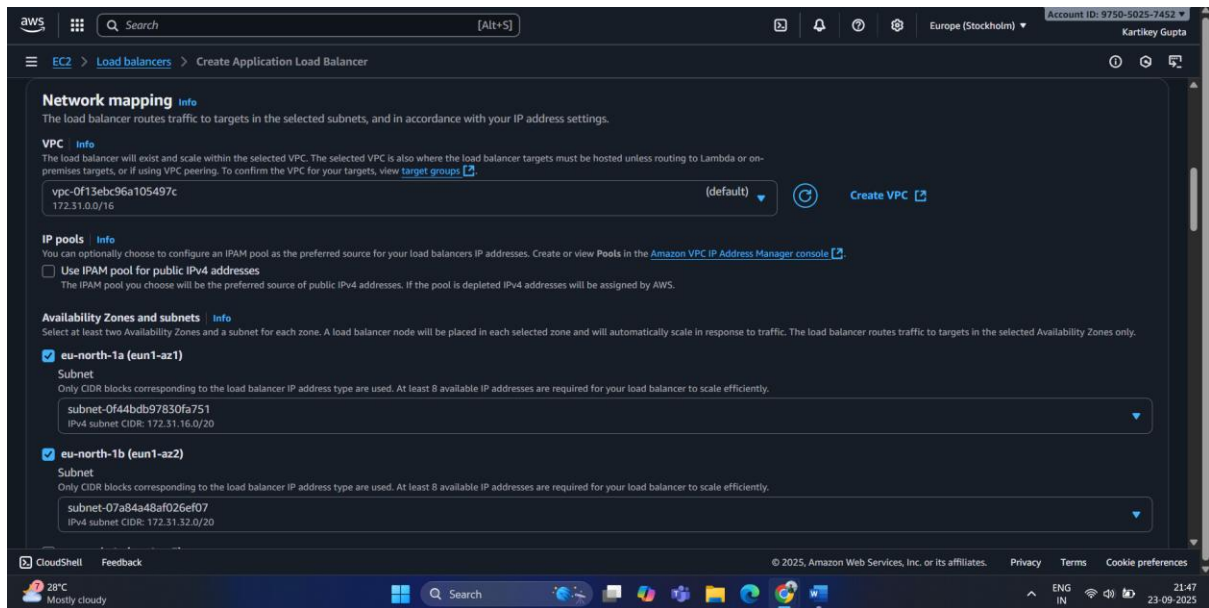
1. Go to **EC2 → Load Balancers → Create Load Balancer → Application Load Balancer**.
2. Configure:
 - **Name:** MyAppALB
 - **Scheme:** Internet-facing
 - **IP address type:** IPv4
 - **Listeners:** HTTP (80)
3. Select **Availability Zones:** choose at least 2 public subnets in different AZs.





1. Configure:



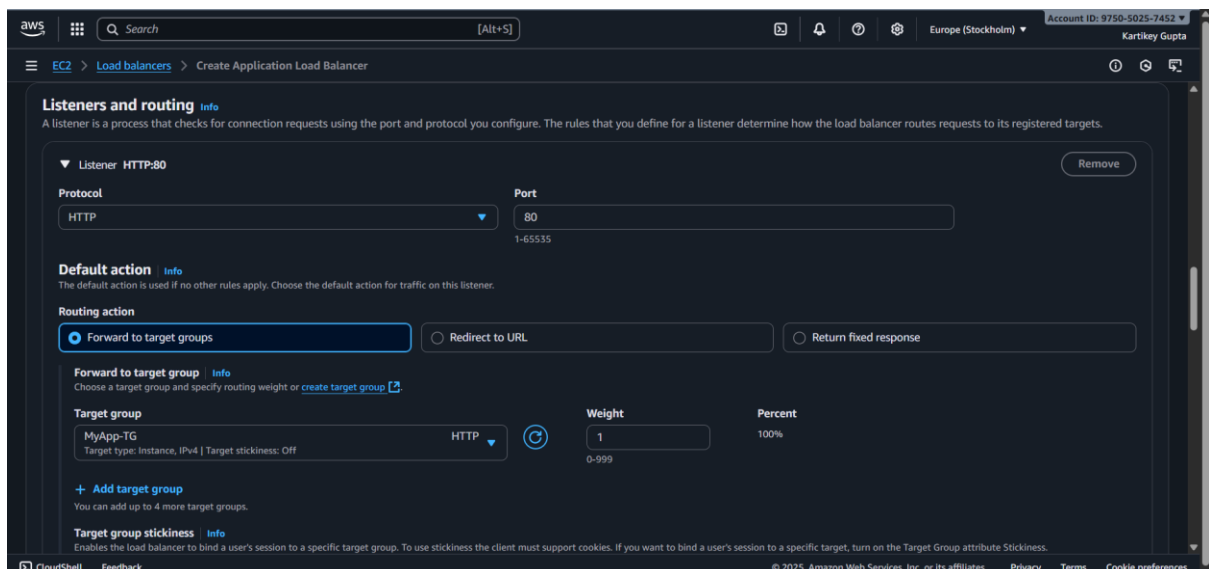


Step 3: Configure Security Group

1. Either select an existing SG or create a new one.
2. **Inbound rule:** HTTP, port 80, source 0.0.0.0/0
3. Outbound can remain default (All traffic).

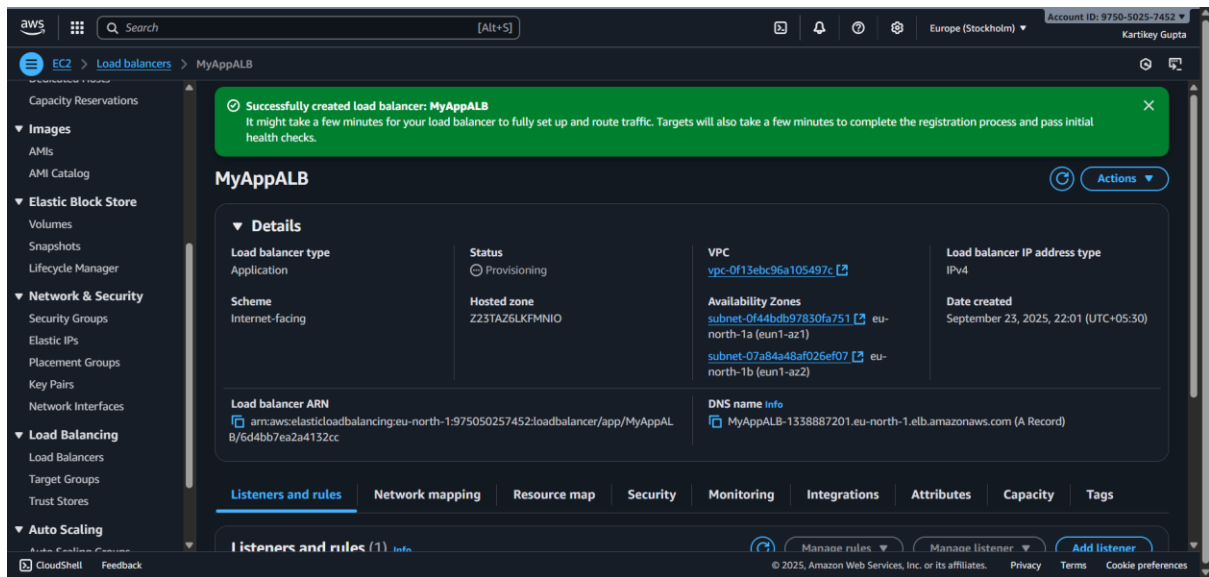
Step 4: Attach Target Group

1. Under **Listener** → **Default action**, select **Forward to** → **Your Target Group** (e.g., MyApp-TG).



Step 5: Review and Create

- Review all settings and click **Create Load Balancer**.
- Note the **ALB DNS name** — this is what users will access.



Step (iii): Configure Health Checks

Step 1: Open Target Groups

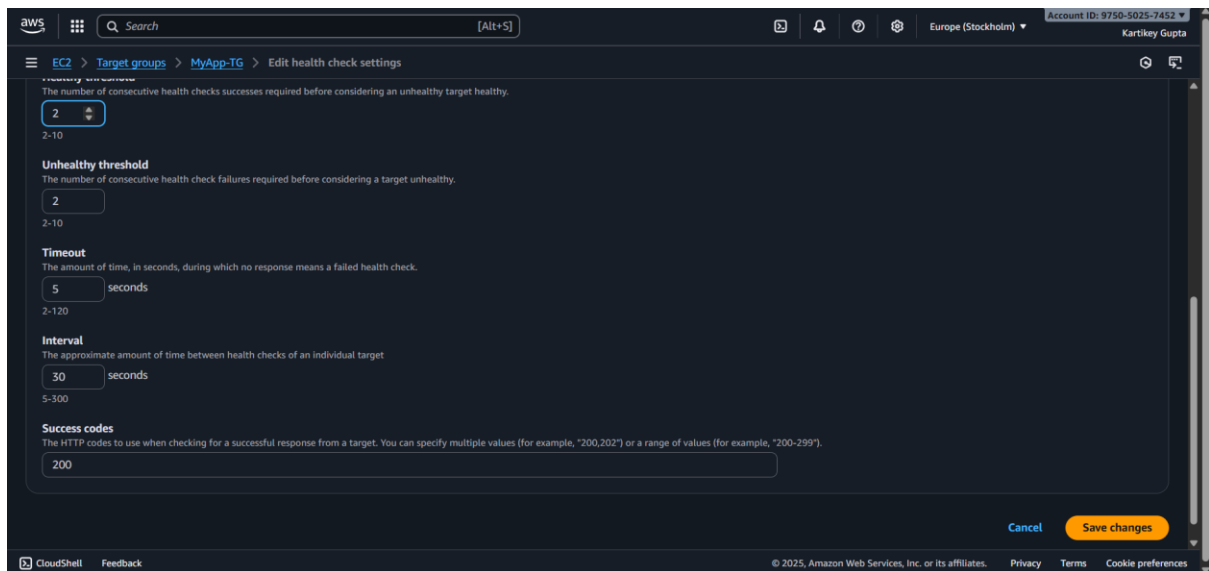
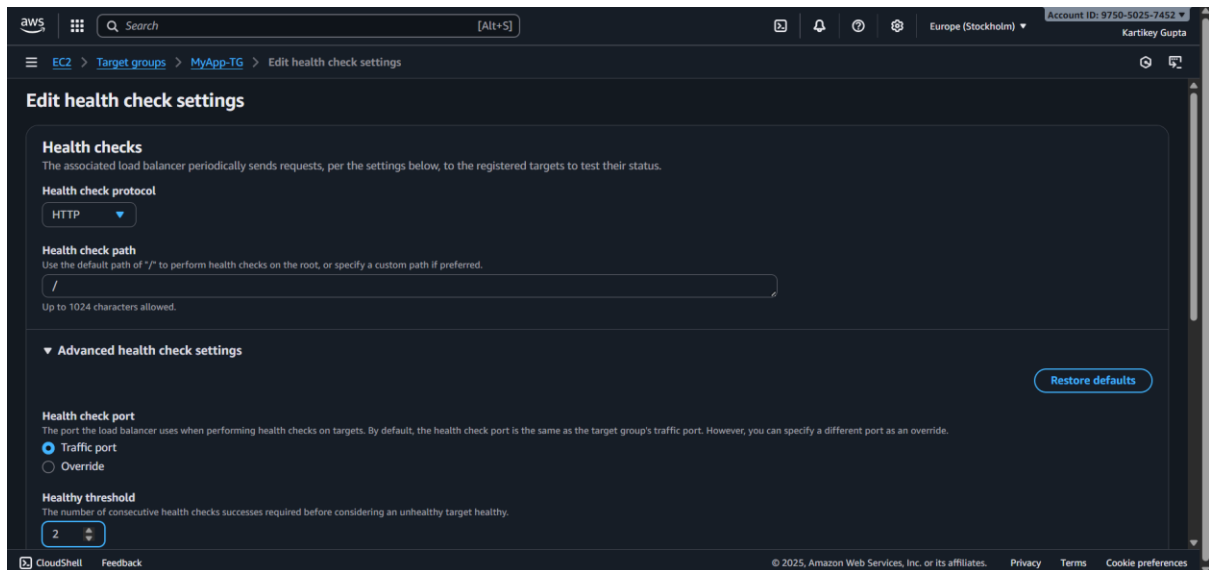
1. Go to **EC2** → **Target Groups** in the AWS console.
2. Select the **Target Group** you created (e.g., MyApp-TG).

Step 2: Go to Health Checks

1. In the Target Group details, click the **Health checks** tab.

Step 3: Configure Health Check Settings

- **Protocol:** HTTP
- **Path:** / (or /health if you have a dedicated health endpoint)
- **Healthy threshold:** 2 → instance must pass 2 consecutive checks to be marked healthy
- **Unhealthy threshold:** 2 → instance must fail 2 consecutive checks to be marked unhealthy
- **Timeout:** 5 seconds → how long to wait for a response
- **Interval:** 30 seconds → time between checks



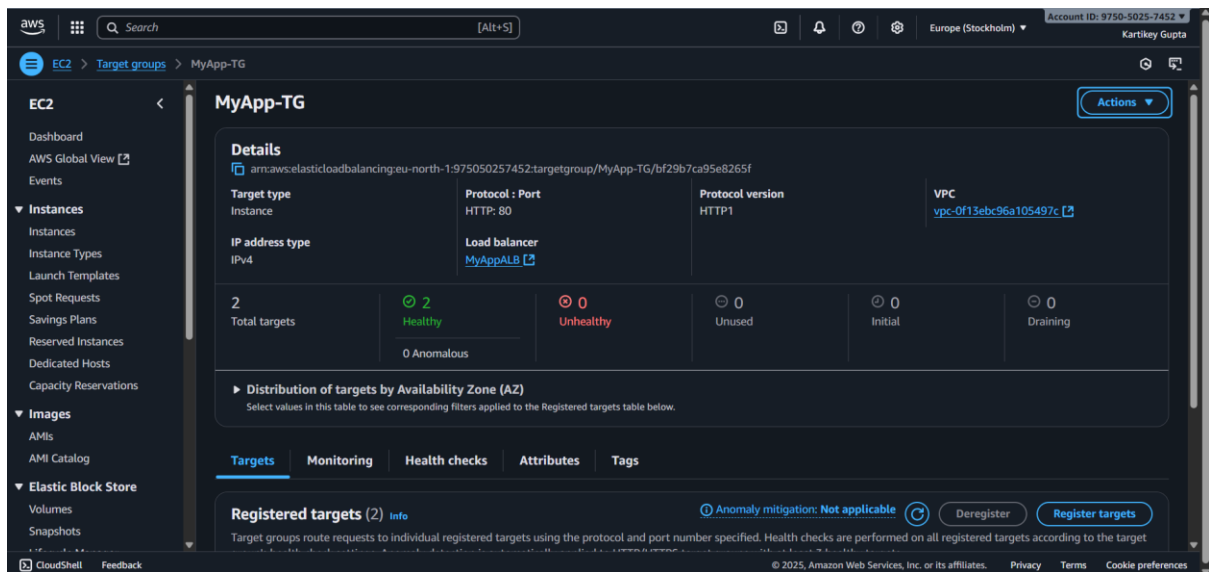
Step 4: Save

- Click **Save** to apply health check settings.

Step 5: Verify

- Go back to **Targets** tab in the Target Group.
- After a few seconds/minutes, the **Status** column should show **healthy** for your EC2 instances.
- If an instance fails, check:
 - Security group allows inbound HTTP from ALB

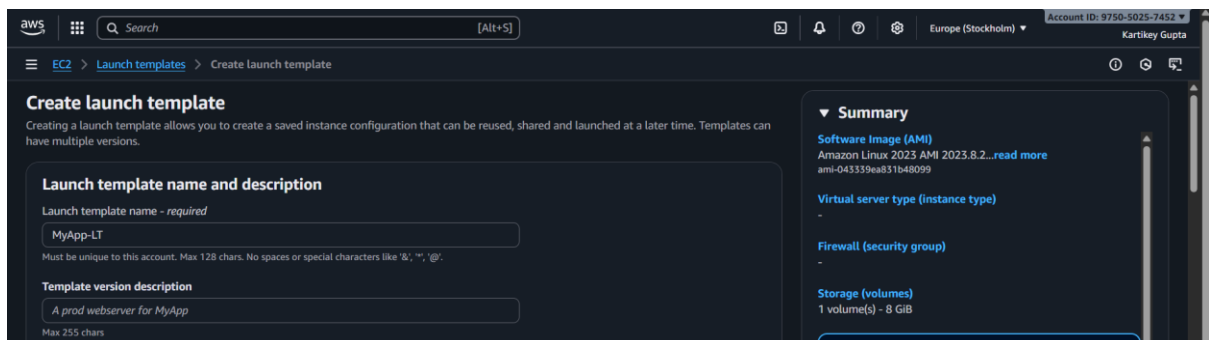
- Web server is running and responding at the path you configured

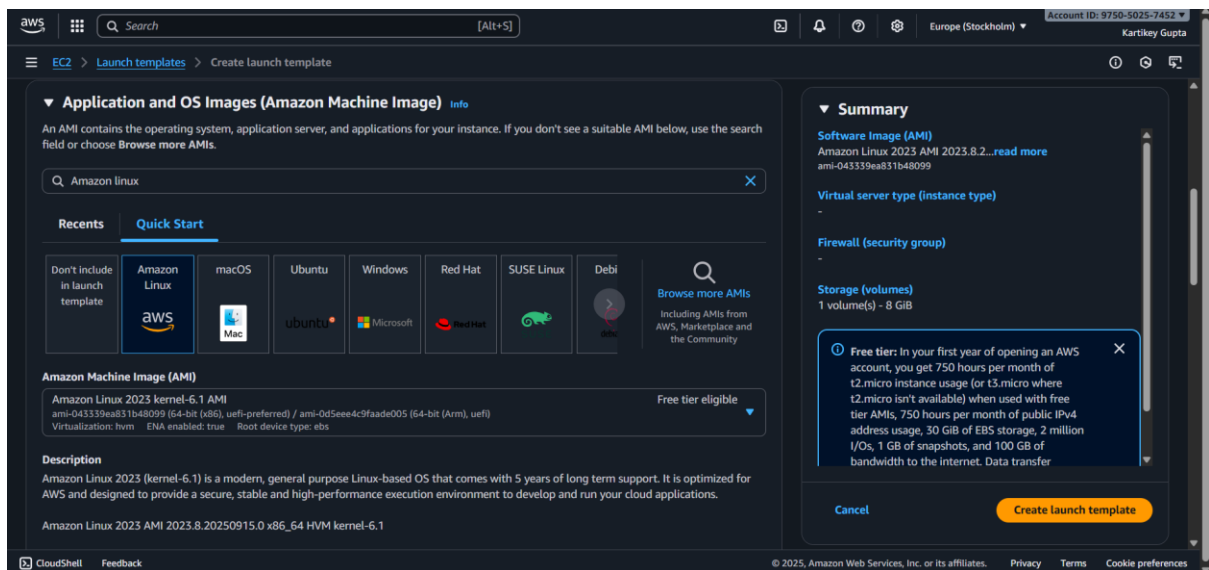


Step (iv): Enable Auto Scaling

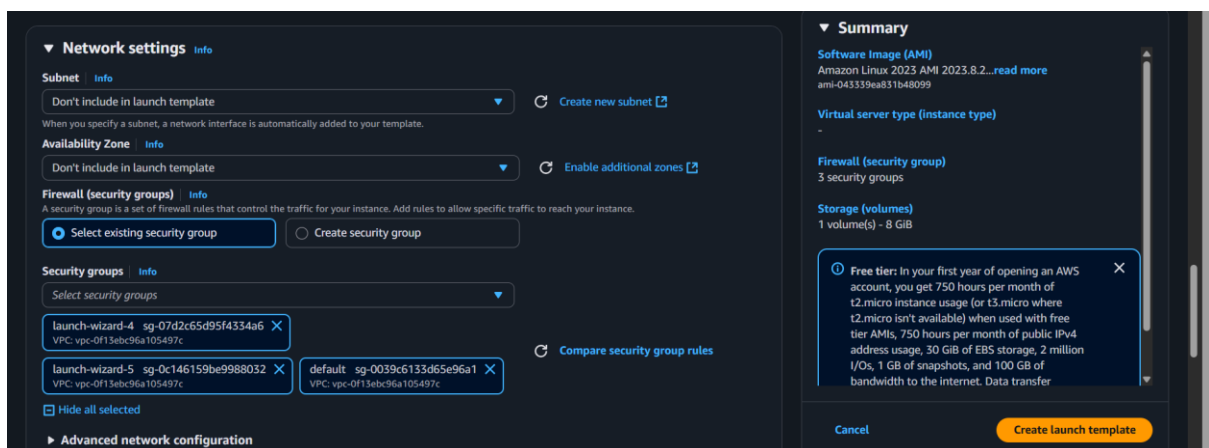
Step 1: Prepare a Launch Template

1. Go to **EC2** → **Launch Templates** → **Create Launch Template**.
2. Configure template:
 - **Name:** MyApp-LT
 - **AMI:** Use the same AMI as your EC2 instances





- **Instance type:** e.g., t2.micro
- **Key pair:** Your existing key
- **Security group:** Same as your running EC2 (allow HTTP 80 from ALB SG)



- **User data (optional):** Script to auto-install Apache/web server on launch:

```
#!/bin/bash
```

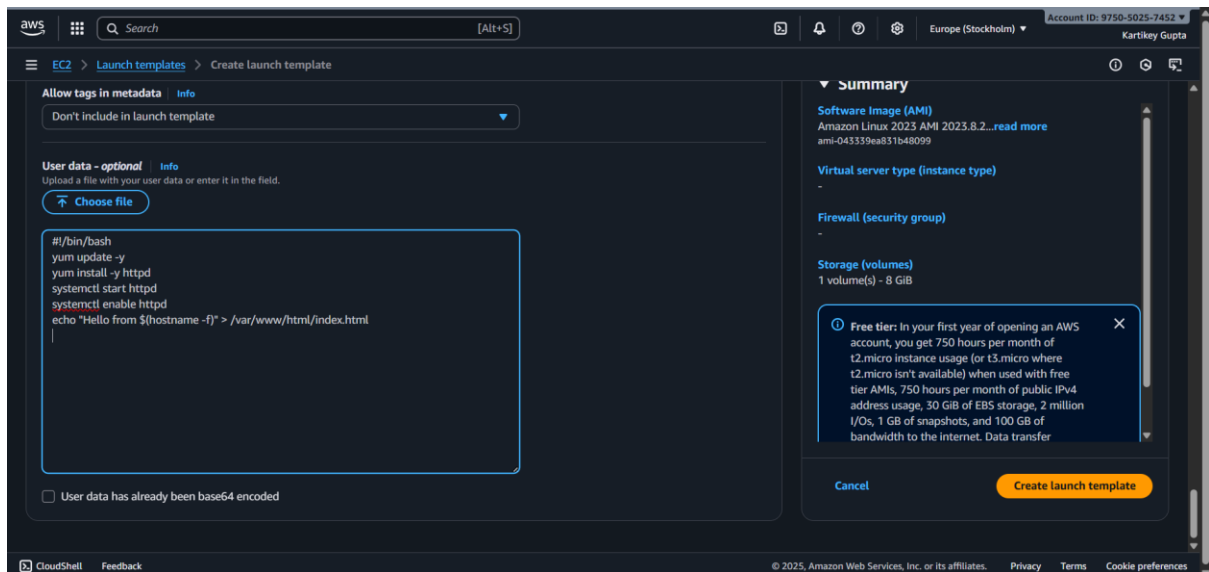
```
yum update -y
```

```
yum install -y httpd
```

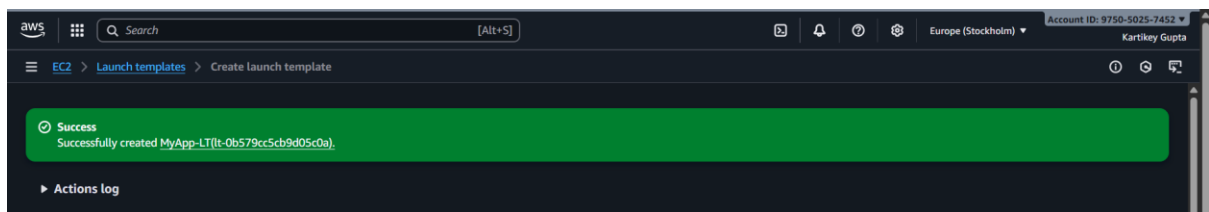
```
systemctl start httpd
```

```
systemctl enable httpd
```

```
echo "Hello from $(hostname -f)" > /var/www/html/index.html
```

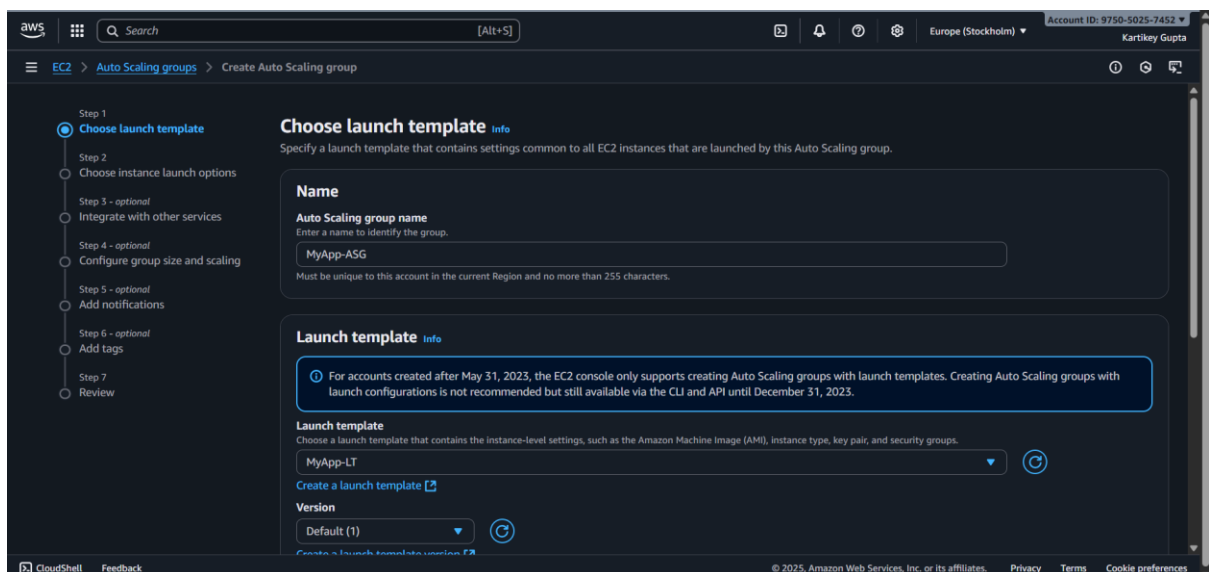


3. Save the Launch Template.



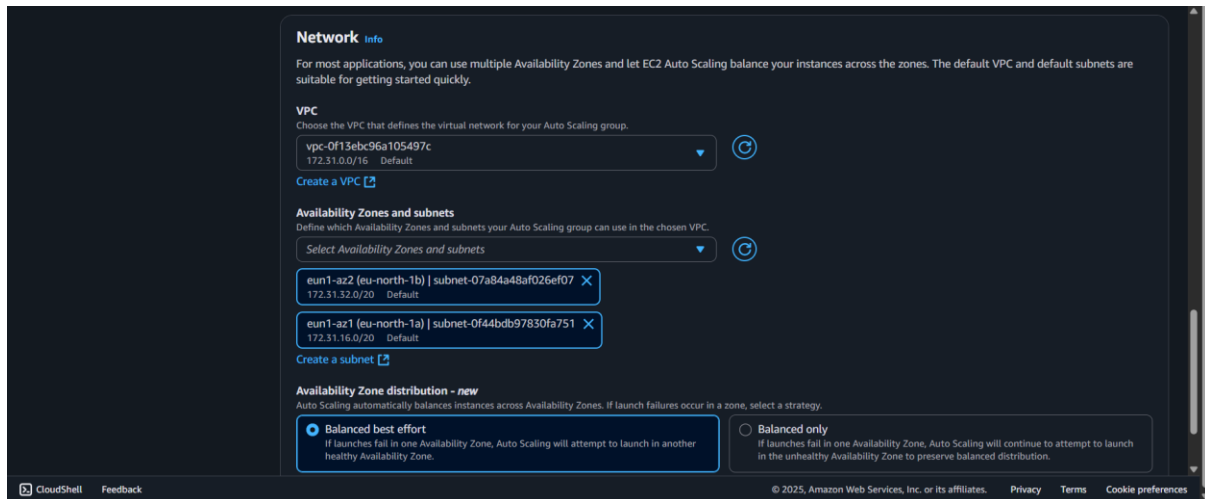
Step 2: Create Auto Scaling Group

1. Go to **EC2 → Auto Scaling Groups → Create Auto Scaling Group**.
2. **Select launch template: MyApp-LT**
3. **Name: MyApp-ASG**



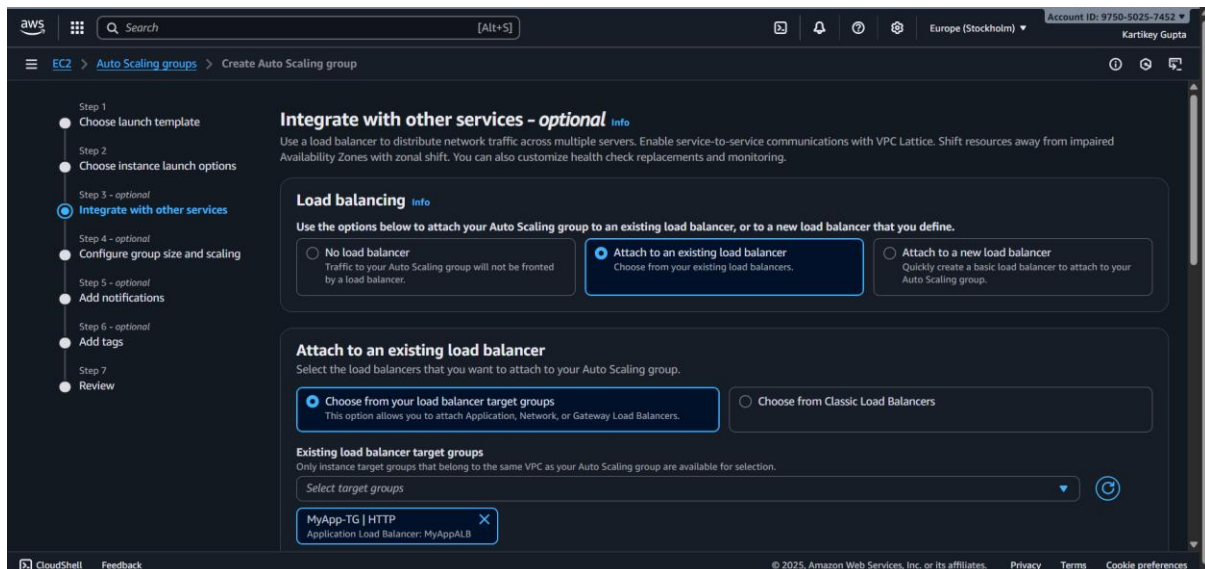
Step 3: Configure Network

1. **VPC:** Same as your instances/ALB
2. **Subnets:** Select at least **2 AZs** for high availability



Step 4: Attach to Target Group

1. Select “**Attach to an existing load balancer**”
2. Choose your **ALB Target Group** (MyApp-TG)



Step 5: Configure Group Size

- **Desired capacity:** 2 → initially 2 instances

- **Minimum: 2** → ensures at least 2 instances are always running
- **Maximum: 5** → allows scaling up to 5 instances if needed

The screenshot shows the 'Configure group size and scaling' step in the AWS Management Console. The left sidebar lists steps from 'Choose launch template' to 'Review', with 'Configure group size and scaling' selected. The main content area is titled 'Configure group size and scaling - optional' and includes sections for 'Group size', 'Desired capacity type', 'Desired capacity', and 'Scaling limits'. The 'Desired capacity' is set to 2. The 'Scaling limits' section shows 'Min desired capacity' as 2 and 'Max desired capacity' as 5.

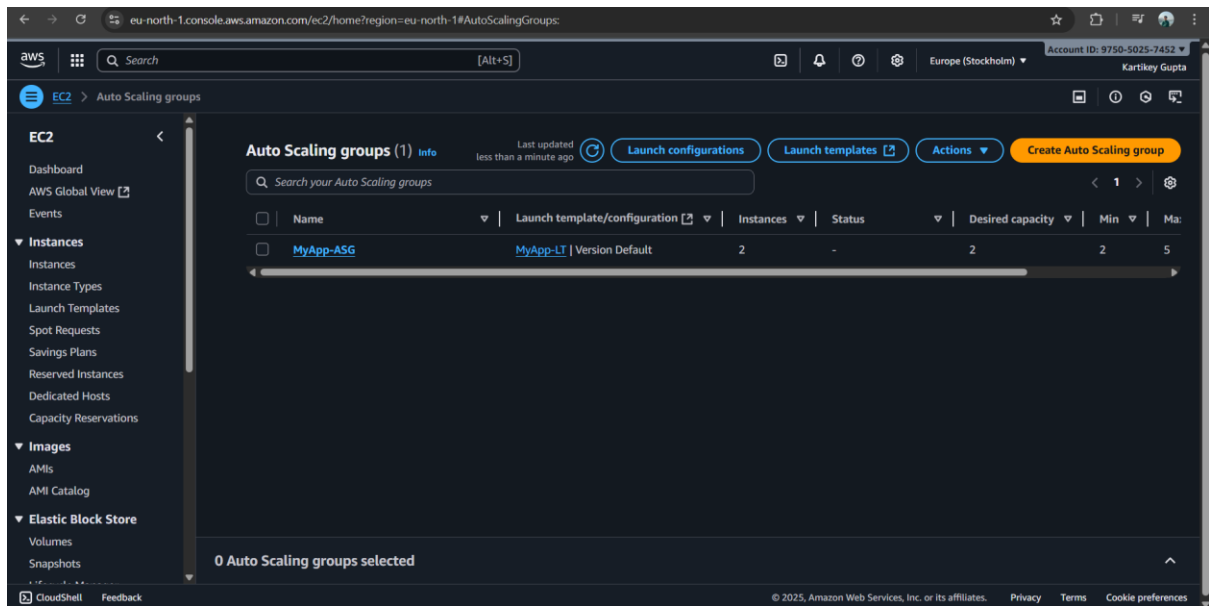
Step 6: Add Scaling Policies

1. Target tracking / simple scaling
2. Example policy:
 - **Scale out:** CPU > 70% → add instance(s)

The screenshot shows the 'Automatic scaling' step in the AWS Management Console. The left sidebar is the same as the previous screenshot. The main content area is titled 'Automatic scaling - optional' and includes sections for 'Choose whether to use a target tracking policy', 'Scaling policy name', 'Metric type', 'Target value', and 'Instance warmup'. The 'Target tracking scaling policy' is selected. The 'Scaling policy name' is 'Target Tracking Policy'. The 'Metric type' is 'Average CPU utilization'. The 'Target value' is 70. The 'Instance warmup' is set to 300 seconds.

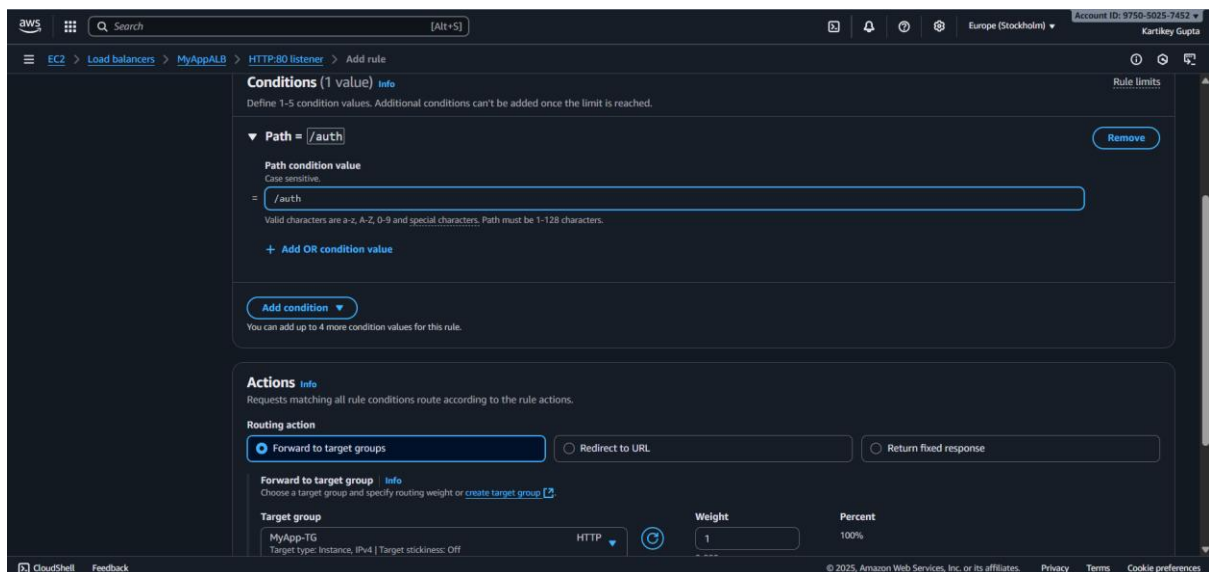
Step 7: Review and Create

- Review all settings → click **Create Auto Scaling Group**
- Auto Scaling will now automatically launch 2 instances initially, register them with ALB, and scale based on traffic.



(v) Configure Path-Based Routing + Route 53

1. Go to **ALB** → **Listeners** → **View/Edit rules**.
2. Add rules:
 - If path is `/auth` → forward to **Target Group: AuthService**.
 - If path is `/order` → forward to **Target Group: OrderService**.
3. Ensure you created **separate EC2 groups** (Auth, Order) running respective services.
4. Save listener rules.



EC2

Load balancers

MyAppALB

HTTP:80 listener

Add rule

▼ Path = /order

Path condition value

Case sensitive.

= /order

Valid characters are a-z, A-Z, 0-9 and special characters. Path must be 1-128 characters.

+ Add OR condition value

Add condition ▼

You can add up to 4 more condition values for this rule.

Actions

Info

Requests matching all rule conditions route according to the rule actions.

Routing action

Forward to target groups

Redirect to URL

Return fixed response

Forward to target group

Info

Choose a target group and specify routing weight or create target group

Target group

MyApp-TG

Target type: Instance, IPv4 | Target stickiness: Off

HTTP

Weight

1

0-999

Percent

100%

+ Add target group

CloudShell

Feedback

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EC2

Load balancers

MyAppALB

HTTP:80 listener

MyAppALB

Forward to target group

MyApp-TG 1 (100%)

Target group stickiness: Off

Listener ARN

arn:aws:elasticloadbalancing:eu-north-1:975050257452:listener/app/MyAppALB/6d4bb7ea2a4132cc/ea37fbdacccc0b

Rules

Attributes

Tags

Listener rules (3)

Info

Rule limits

Actions

Add rule

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

Filter rules

	Priority	Name tag	Conditions (If)	Actions (Then)	ARN	Tags	Actions
☐	1	-	Path = /order	Forward to target group MyApp-TG 1 (100%) Target group stickiness: Off	ARN	0 tags	
☐	2	-	Path = /auth	Forward to target group MyApp-TG 1 (100%) Target group stickiness: Off	ARN	0 tags	
☐	Last (default)	Default	If no other rule applies	Forward to target group MyApp-TG 1 (100%) Target group stickiness: Off	ARN	0 tags	

CloudShell

Feedback

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